

VPC And Peering

Problem Statement:

You work for XYZ Corporation and based on the expansion requirements of your corporation you have been asked to create and set up a distinct Amazon VPC for the production and development team. You are expected to perform the following tasks for the respective VPCs.

Production Network:

- 1. Design and build a 4-tier architecture. 2. Create 5 subnets out of which 4 should be private named app1, app2, dbcache and db and one should be public, named web**
- 3. Launch instances in all subnets and name them as per the subnet that they have been launched in.**
- 4. Allow dbcache instance and app1 subnet to send internet requests**
- 5. Manage security group and NACL's**

Development Network:

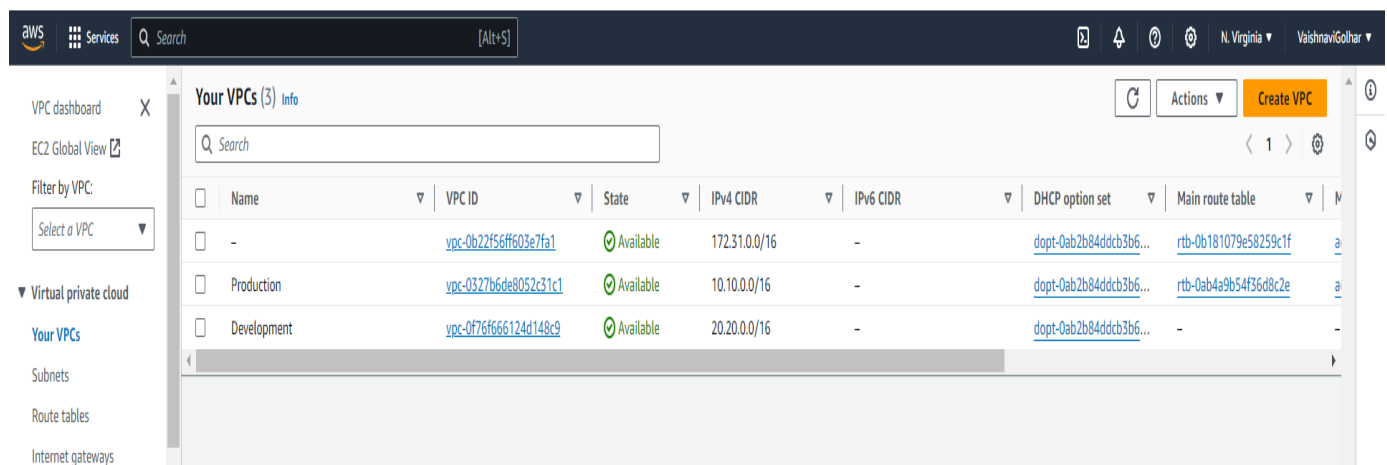
- 1. Design and build 2-tier architecture with two subnets named web and db and launch instances in both subnets and name them as per the subnet names.**
- 2. Make sure only the web subnet can send internet requests.**

3. Create peering connection between production network and development network.
4. Setup connection between db subnets of both production network and development network respectively.

Solution:

For Production Network

1. Open the AWS management Console.
2. Check the region
3. Now create 2 VPC's 1) Production 2) Development



4. Create 5 subnet in the Production VPC
 - a) App-1
 - b) App-2
 - c) DB
 - d) DB-CACHE
 - e) WEB

Subnets (5) Info

Find resources by attribute or tag

Filter by VPC: Clear filters

	Name	Subnet ID	State	VPC	IPv4 CIDR	IPv6 CIDR	Available IPv4 addresses
☐	APP-2	subnet-04d7e87751d2ac3b6	Available	vpc-0327b6de8052c31c1 Prod...	10.10.2.0/24	-	251
☐	DB-CACHE	subnet-0d5b55b68a859d099	Available	vpc-0327b6de8052c31c1 Prod...	10.10.3.0/24	-	251
☐	DB	subnet-0c799334c98542344	Available	vpc-0327b6de8052c31c1 Prod...	10.10.4.0/24	-	251
☐	WEB	subnet-08ea3d1cf9db288c3	Available	vpc-0327b6de8052c31c1 Prod...	10.10.5.0/24	-	251
☐	APP-1	subnet-059ea29c4df9e3a7	Available	vpc-0327b6de8052c31c1 Prod...	10.10.0.0/24	-	251

5. Make WEB subnet public by assign Public Ip

VPC > Subnets > subnet-08ea3d1cf9db288c3 > Edit subnet settings

Edit subnet settings

Subnet

Subnet ID: subnet-08ea3d1cf9db288c3 Name: WEB

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address

☐ Enable auto-assign customer-owned IPv4 address

6. Create internet Gateway with the name Production-IGW

VPC > Internet gateways > Create internet gateway

Create internet gateway

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag: Production-IGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key: Name Value - optional: Production-IGW

Buttons: Add new tag, Cancel, Create internet gateway

7. Attach this Production-IGW to Production VPC

The screenshot shows the AWS Management Console interface for attaching an Internet Gateway to a VPC. The breadcrumb navigation indicates the path: VPC > Internet gateways > Attach to VPC (igw-07f8a961f1f1358f3). The main heading is 'Attach to VPC (igw-07f8a961f1f1358f3)' with an 'Info' link. Below this, a section titled 'VPC' explains that attaching an internet gateway enables the VPC to communicate with the internet. A list of 'Available VPCs' is shown, with a search bar containing 'vpc-0327b6de8052c31c1'. The search results list 'vpc-0327b6de8052c31c1 - Production' as the selected option. Other VPCs listed include 'vpc-0b22f56ff603e7fa1' and 'vpc-07f6f666124d148c9 - Development'. An orange 'Attach internet gateway' button is visible at the bottom right of the VPC list.

8. Create Route table with the name Production-RT in VPC of Production.

The screenshot shows the AWS Management Console interface for creating a new route table. The breadcrumb navigation indicates the path: VPC > Route tables > Create route table. The main heading is 'Create route table' with an 'Info' link. Below this, a section titled 'Route table settings' contains a 'Name - optional' field with the value 'Production-RT' and a 'VPC' dropdown menu showing 'vpc-0327b6de8052c31c1 (Production)'. A 'Tags' section is also visible, showing a key-value pair 'Name' with value 'Production-RT'. At the bottom, there are 'Cancel' and 'Create route table' buttons.

9. Edit routes with destination as internet (0.0.0.0/0) to target internet gateway to Production (Production-IGW).

Destination

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No
0.0.0.0/0	Internet Gateway	Active	No

Buttons: Add route, Cancel, Preview, Save changes

10. To get internet to public subnet (WEB) make subnet Association with WEB.

Available subnets (1/5)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
APP-2	subnet-04d7e87751d2ac3b6	10.10.2.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
DB-CACHE	subnet-0d5b55b68a859d099	10.10.3.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
DB	subnet-0c799334c98542344	10.10.4.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
☒ WEB	subnet-08ea3d1cf9db288c3	10.10.5.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
APP-1	subnet-059eea9c4dfd9e3a7	10.10.0.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)

Selected subnets

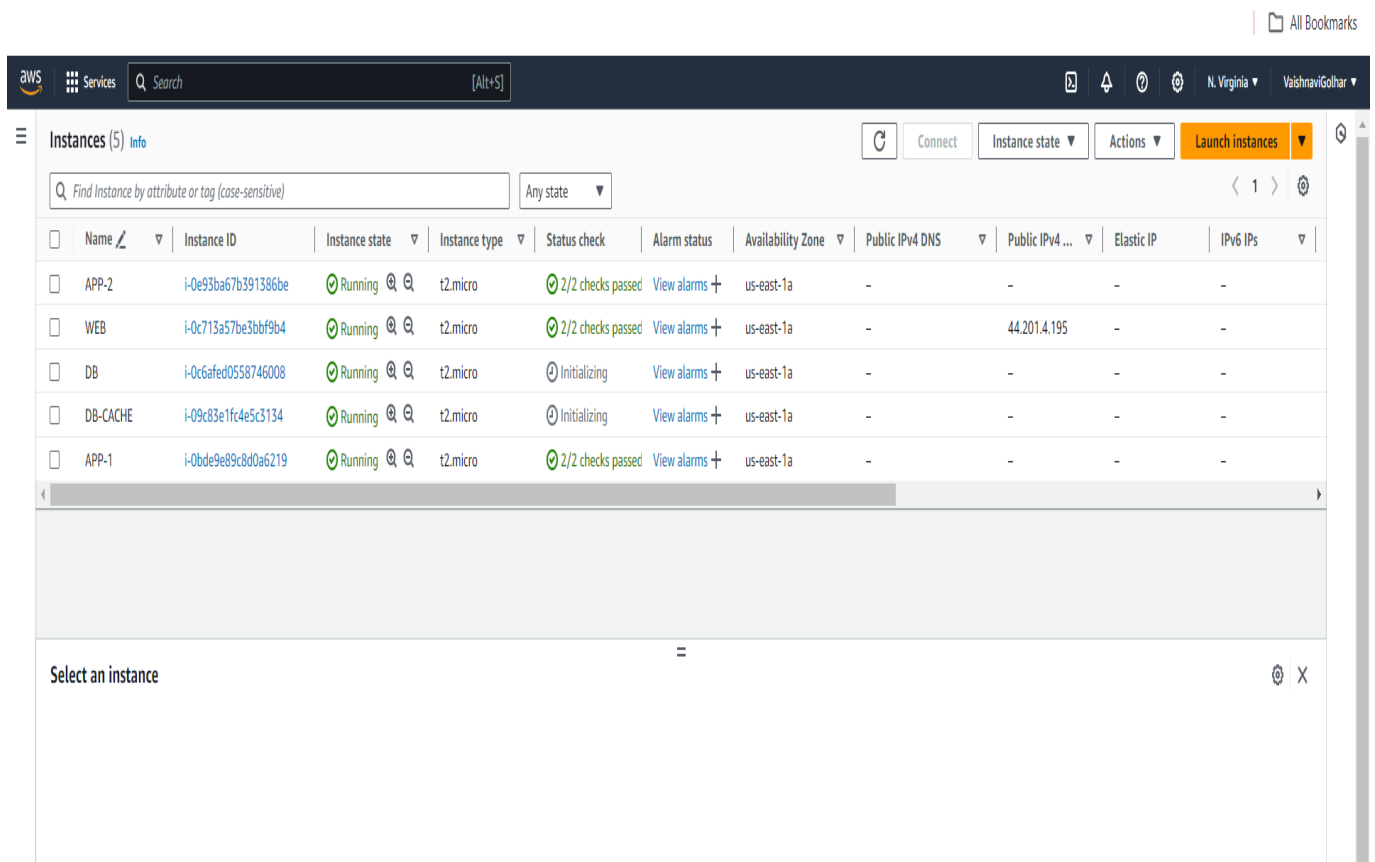
subnet-08ea3d1cf9db288c3 / WEB

Buttons: Cancel, Save associations

11. Now launch 5 EC2 instance in the Production VPC

a) App-1 b) APP-2 c) DB d) DB-CACHE e) WEB

with all ubuntu instances in their respectively subnets with default security groups.



The screenshot displays the AWS Management Console's 'Instances' page. It shows a list of five EC2 instances, all of which are 'Running'. The instances are named APP-2, WEB, DB, DB-CACHE, and APP-1. They are all using the 't2.micro' instance type and are located in the 'us-east-1a' availability zone. The 'WEB' instance has a public IPv4 address of 44.201.4.195. The 'DB' and 'DB-CACHE' instances are in the 'Initializing' state. The 'APP-1' instance is also in the 'Running' state. The console interface includes a search bar, a filter dropdown, and a table of instance details.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
APP-2	i-0e93ba67b391386be	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-	-	-
WEB	i-0c713a57be3bbf9b4	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	44.201.4.195	-	-
DB	i-0c6afed058746008	Running	t2.micro	Initializing	View alarms +	us-east-1a	-	-	-	-
DB-CACHE	i-09c83e1fc4e5c3134	Running	t2.micro	Initializing	View alarms +	us-east-1a	-	-	-	-
APP-1	i-0bde9e89c8d0a6219	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-	-	-

12. Now check the internet connecting in the public subnet (WEB). So connect instance and ping googl.com

Also update with commands `sudo nano FirstKey.pem`, `sudo chmod 400 FirstKey.pem`, `sudo ssh -i ubuntu "FirstKey.pem" ubuntu@44.201.4.195`

```
aws Services Search [Alt+S]
ubuntu@ip-10-10-5-169:~$ sudo nano FirstKey.pem
ubuntu@ip-10-10-5-169:~$ ls
FirstKey.pem
ubuntu@ip-10-10-5-169:~$ sudo chmod 400 FirstKey.pem
ubuntu@ip-10-10-5-169:~$ ssh -i "FirstKey.pem" ubuntu@44.201.4.195
The authenticity of host '44.201.4.195 (44.201.4.195)' can't be established.
ED25519 key fingerprint is SHA256:ah1BUjTVzERQfam8OT7JCYgk/3EbRqu9eubXGTMk43A.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.201.4.195' (ED25519) to the list of known hosts.
Load key "FirstKey.pem": Permission denied
ubuntu@44.201.4.195: Permission denied (publickey).
ubuntu@ip-10-10-5-169:~$ ping google.com
PING google.com (172.253.122.139) 56(84) bytes of data.
64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=1 ttl=101 time=2.36 ms
64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=2 ttl=101 time=2.50 ms
^X64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=3 ttl=101 time=2.44 ms
^C

i-0c713a57be3bbf9b4 (WEB)
PublicIPs: 44.201.4.195 PrivateIPs: 10.10.5.169
```

13. As per the question allow DB-CACHE instance and App-1 Subnet to send internet requests. For that create NAT gateway with the name Production-NAT in the WEB subnet and connectivity type public with an elastic IP allocation to it.

aws Services Search [Alt+S]

Elastic IP address 54.210.193.242 (eipalloc-0b77b8d75b347a40e) allocated.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Subnet
Select a subnet in which to create the NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

► Additional settings [Info](#)

14. Create another route table with a name NAT-PRODUCTION and edit route for make connection to public (WEB) subnet to private subnets (APP-1 and DB-CACHE) to pass internet requests. As Destination 0.0.0.0/0 to target NAT Gateway of NAT-PRODUCTION.

The screenshot shows the 'Create route table' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Route tables > Create route table'. The page title is 'Create route table' with an 'Info' link. A descriptive text states: 'A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.'

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
NAT-PRODUCTION

VPC
The VPC to use for this route table.
vpc-0327b6de8052c31c1 (Production)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q NAT-PRODUCTION X Remove

Add new tag

You can add up to 50 tags per resource.

The screenshot shows the 'Edit routes' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Route tables > rtb-026df6ed92c83c983 > Edit routes'. The page title is 'Edit routes'.

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No
Q 0.0.0.0/0 X	NAT Gateway	-	No
	Q nat- X		
	Use: "nat-"		
	nat-0b7b84dd8bc85d255 (Production-Nat)		

Add route

Remove

Cancel Preview Save changes

15. Make subnet association with APP-1 AND DB-CACHE for internet request.

The screenshot shows the 'Edit subnet associations' page in the AWS Management Console. The breadcrumb trail is: VPC > Route tables > rtb-026df6ed92c83c983 > Edit subnet associations. The page title is 'Edit subnet associations' with a subtitle 'Change which subnets are associated with this route table.' Below this is a section 'Available subnets (2/5)' with a search bar 'Filter subnet associations'. A table lists available subnets with columns: Name, Subnet ID, IPv4 CIDR, IPv6 CIDR, and Route table ID. The subnets listed are APP-2, DB-CACHE (selected), DB, WEB, and APP-1 (selected). Below the table is a 'Selected subnets' section showing two selected subnets: 'subnet-059eea9c4dfd9e3a7 / APP-1' and 'subnet-0d5b55b68a859d099 / DB-CACHE'. At the bottom right are 'Cancel' and 'Save associations' buttons.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
APP-2	subnet-04d7e87751d2ac3b6	10.10.2.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
☒ DB-CACHE	subnet-0d5b55b68a859d099	10.10.3.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
DB	subnet-0c799334c98542344	10.10.4.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)
WEB	subnet-08ea3d1cf9db288c3	10.10.5.0/24	-	rtb-0e9eeae311bd854a8 / Production...
☒ APP-1	subnet-059eea9c4dfd9e3a7	10.10.0.0/24	-	Main (rtb-0ab4a9b54f36d8c2e)

The screenshot shows the 'Subnet associations' tab for route table rtb-026df6ed92c83c983. A green notification banner at the top states: 'You have successfully updated subnet associations for rtb-026df6ed92c83c983 / NAT-PRODUCTION.' The 'Details' section shows: Route table ID: rtb-026df6ed92c83c983, Main: No, Explicit subnet associations: 2 subnets, Edge associations: -, VPC: vpc-0327b6de8052c31c1 | Production, Owner ID: 161681788148. Below this is a tabbed interface with 'Subnet associations' selected. The 'Explicit subnet associations (2)' section shows a table with columns: Name, Subnet ID, IPv4 CIDR, and IPv6 CIDR. The subnets listed are DB-CACHE and APP-1.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
DB-CACHE	subnet-0d5b55b68a859d099	10.10.3.0/24	-
APP-1	subnet-059eea9c4dfd9e3a7	10.10.0.0/24	-

16. To check internet request from public subnet of WEB instance to the APP-1 and DB-CACHE instances.

Now check the internet connecting in the public subnet (WEB). So, connect instance and update with commands

sudo nano FirstKey.pem, sudo chmod 400 FirstKey.pem,

sudo ssh -i "FirstKey.pem" [ubuntu@10.10.3.107](#)

sudo apt update and ping google.com

```
aws Services Search [Alt+S] N. Virginia VaishnaviGolhar
ubuntu@ip-10-10-5-169:~$ chmod 400 "FirstKey.pem"
chmod: changing permissions of 'FirstKey.pem': Operation not permitted
ubuntu@ip-10-10-5-169:~$ sudo chmod 400 "FirstKey.pem"
ubuntu@ip-10-10-5-169:~$ sudo ssh -i "FirstKey.pem" ubuntu@10.10.3.107
The authenticity of host '10.10.3.107 (10.10.3.107)' can't be established.
ED25519 key fingerprint is SHA256:LWqjvTR5JaR4tkRJyjq0uooJypJxmNfcM71f7OX3N8c.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.3.107' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.2.0-1017-aws x86_64)

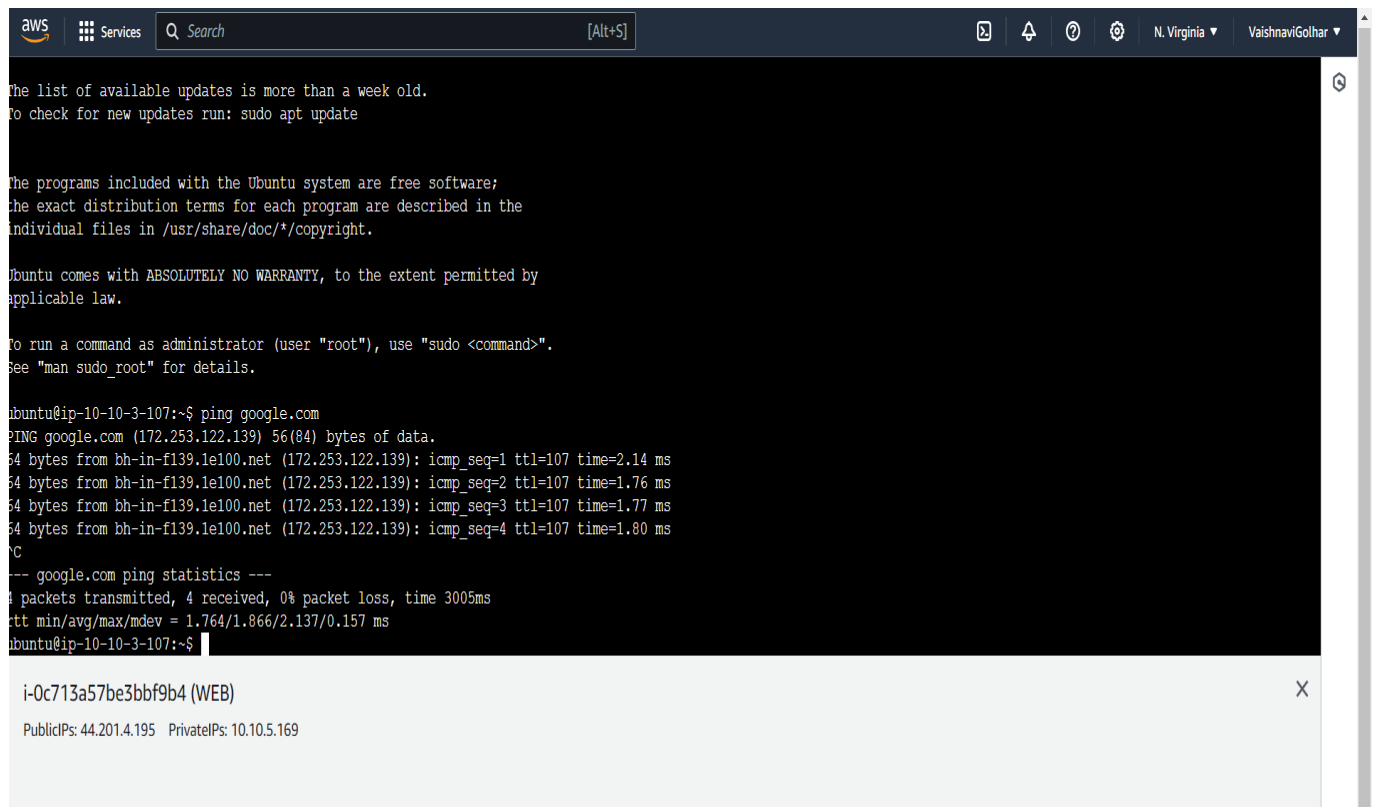
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

System information as of Thu Feb 15 19:42:04 UTC 2024

System load:  0.0          Processes:      97
Usage of /:   20.5% of 7.57GB Users logged in: 0
Memory usage: 20%         IPv4 address for eth0: 10.10.3.107
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
```



The screenshot shows an AWS console terminal window. The top bar includes the AWS logo, 'Services', a search bar, and the user's location 'N. Virginia' and name 'VaishnaviGolhar'. The terminal output displays Ubuntu system information, including update status, program licensing, warranty, and usage instructions. A 'ping google.com' command is executed, showing four successful pings with response times between 1.76ms and 2.14ms. The terminal window has a title bar with the ID 'i-0c713a57be3bbf9b4 (WEB)' and a close button.

```
aws Services Search [Alt+S] N. Virginia VaishnaviGolhar

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-10-3-107:~$ ping google.com
PING google.com (172.253.122.139) 56(84) bytes of data.
64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=1 ttl=107 time=2.14 ms
64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=2 ttl=107 time=1.76 ms
64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=3 ttl=107 time=1.77 ms
64 bytes from bh-in-f139.1e100.net (172.253.122.139): icmp_seq=4 ttl=107 time=1.80 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.764/1.866/2.137/0.157 ms
ubuntu@ip-10-10-3-107:~$
```

i-0c713a57be3bbf9b4 (WEB) X

PublicIPs: 44.201.4.195 PrivateIPs: 10.10.5.169

For Development Network:

1. Already VPC for development network is created.
2. Now create 2 subnets namely a) WEB-2 b) DB-2 in the Development VPC.

Enable auto- assign IP setting for WEB-2

aws Services Search [Alt+S] N. Virginia VaishnaviGolhar

You have successfully created 2 subnets: subnet-094d9e6451353274e, subnet-0f3679750b5c0f581

Subnets (2) Info

Find resources by attribute or tag

Subnet ID : subnet-094d9e6451353274e Subnet ID : subnet-0f3679750b5c0f581 Clear filters

☐	Name	Subnet ID	State	VPC	IPv4 CIDR	IPv6 CIDR	Available IPv4 addresses
☐	WEB-2	subnet-094d9e6451353274e	Available	vpc-0f76f666124d148c9 Deve...	20.20.1.0/24	-	251
☐	DB-2	subnet-0f3679750b5c0f581	Available	vpc-0f76f666124d148c9 Deve...	20.20.2.0/24	-	251

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

3.Create internet Gateway with a name Development-IGW

aws Services Search [Alt+S] N. Virginia VaishnaviGolhar

VPC > Internet gateways > igw-0d3c3d0af0d5f903a

igw-0d3c3d0af0d5f903a / Development-IGW

Actions

Details Info

Internet gateway ID	State	VPC ID	Owner
igw-0d3c3d0af0d5f903a	Detached	-	161681788148

Tags

Search tags

Manage tags

Key	Value
Name	Development-IGW

Virtual private cloud

Your VPCs

Subnets

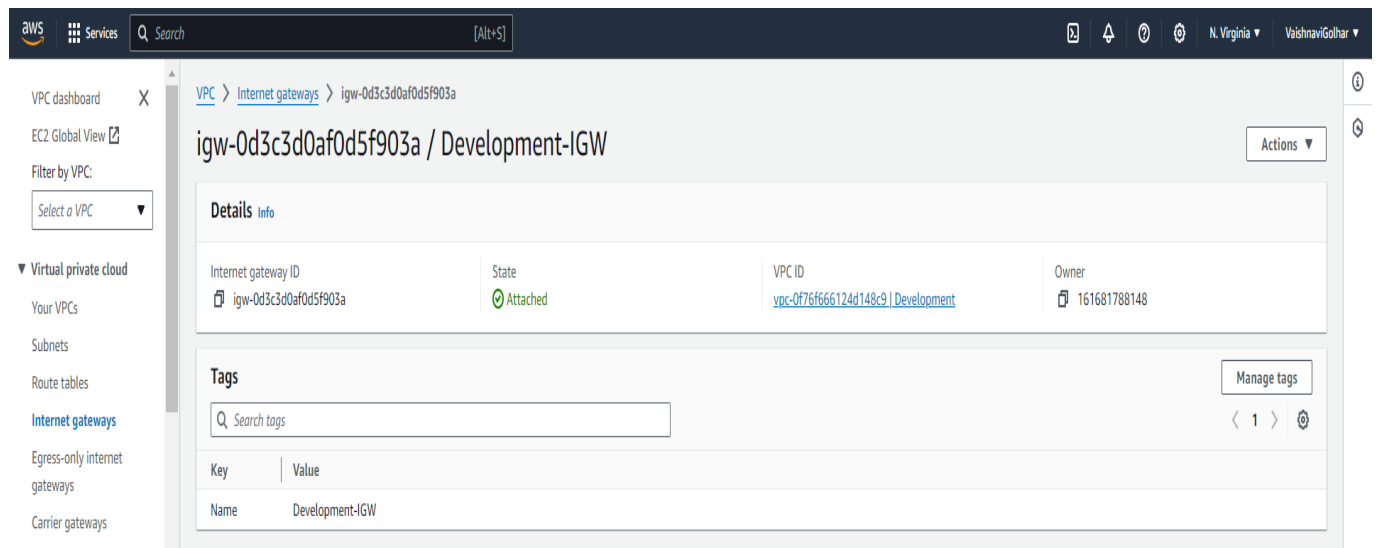
Route tables

Internet gateways

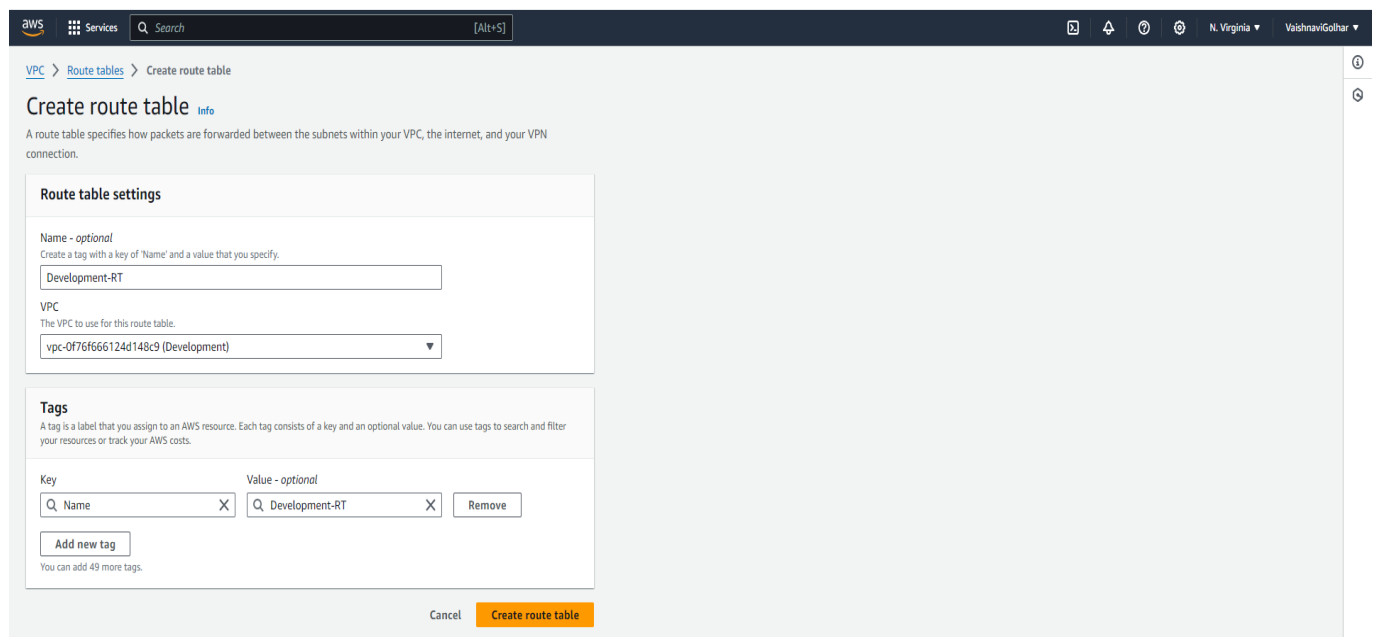
Egress-only internet gateways

Carrier gateways

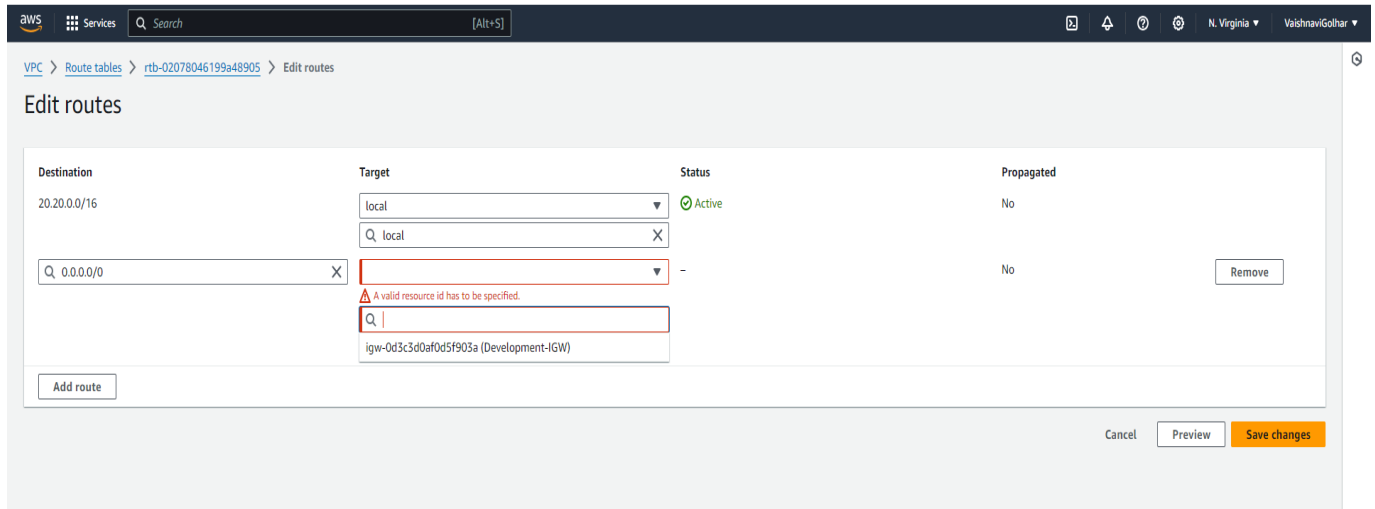
4. Attach this Development-IGW to Development VPC.



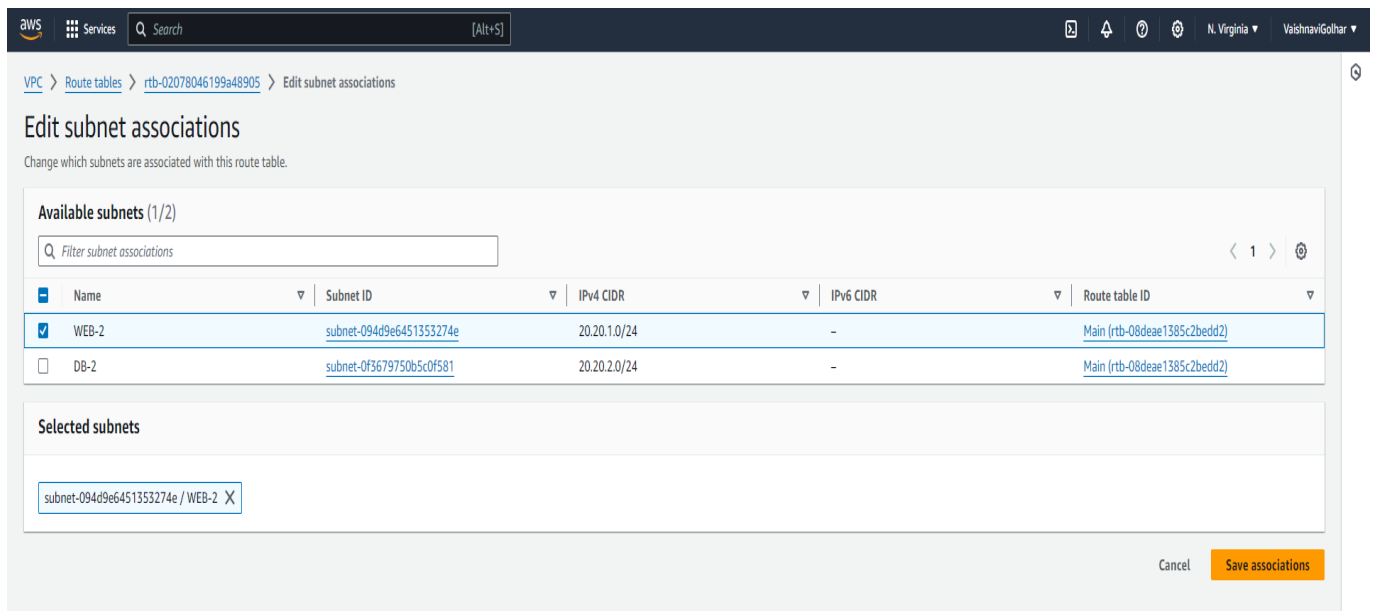
5. Create Route table with the name Development-RT in VPC of Development.



6. Edit routes with destination as internet (0.0.0.0/0) to internet gateway of Development (Development-IGW)

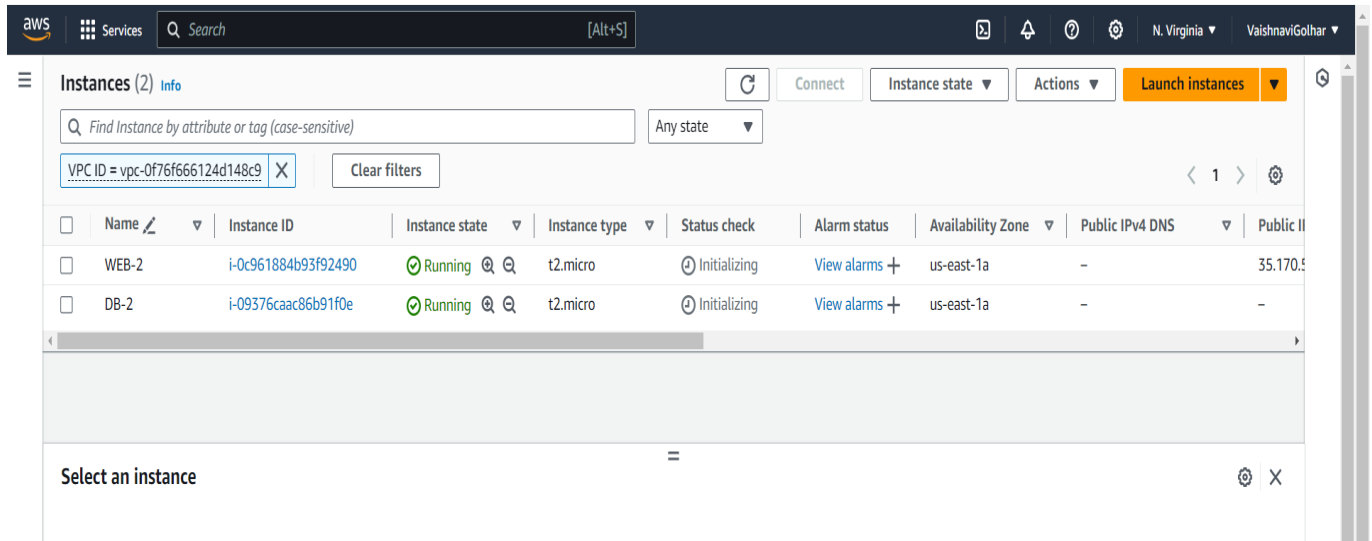


7. To get internet to public subnet (WEB-2) make subnet Association with WEB



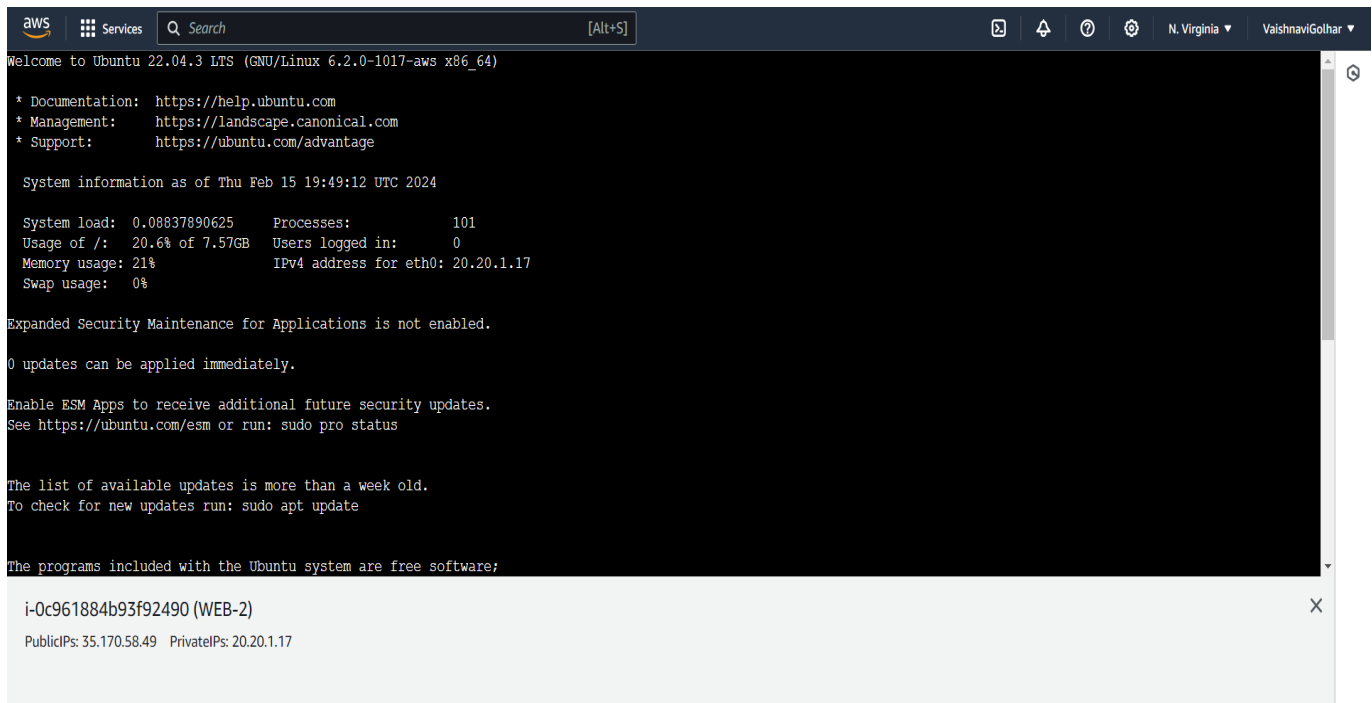
8. Now launch 2 instances in the Development VPC namely.

a) WEB-2 b) DB-2 with all ubuntu instances in their respectively subnets with default SG groups.



9. Now check the internet connectivity in the public subnet (WEB-2).

sudo apt update and ping google.com



```
aws Services Search [Alt+S] N. Virginia VaishnaviGolhar
```

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
ubuntu@ip-20-20-1-17:~$ ping google.com
PING google.com (172.253.115.113) 56(84) bytes of data:
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=1 ttl=50 time=2.58 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=2 ttl=50 time=2.63 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=3 ttl=50 time=2.63 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=4 ttl=50 time=2.72 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=5 ttl=50 time=2.65 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 2.581/2.640/2.716/0.043 ms
ubuntu@ip-20-20-1-17:~$
```

i-0c961884b93f92490 (WEB-2)
PublicIPs: 35.170.58.49 PrivateIPs: 20.20.1.17

Peering Connection:

1. Go to peering connection in VPC dashboard and create peering connection.

aws

Services

Search

[Alt+S]

N. Virginia

VaishnaviGolhar

[VPC](#) > [Peering connections](#) > Create peering connection

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately.

[Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Prod-Dev-Peer

Select a local VPC to peer with

VPC ID (Requester)

vpc-0327b6de8052c31c1 (Production)

VPC CIDRs for vpc-0327b6de8052c31c1 (Production)

CIDR	Status	Status reason
10.10.0.0/16	Associated	-

2. Now create peering connection with the name PROD-DEV-PEER having requester VPC as Production and acceptor VPC as Development.

aws

Services

Search

[Alt+S]

N. Virginia

VaishnaviGolhar

Account

☒ My account

☐ Another account

Region

☒ This Region (us-east-1)

☐ Another Region

VPC ID (Acceptor)

vpc-0f76f666124d148c9 (Development)

VPC CIDRs for vpc-0f76f666124d148c9 (Development)

CIDR	Status	Status reason
20.20.0.0/16	Associated	-

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q Prod-Dev-Peer X

Remove

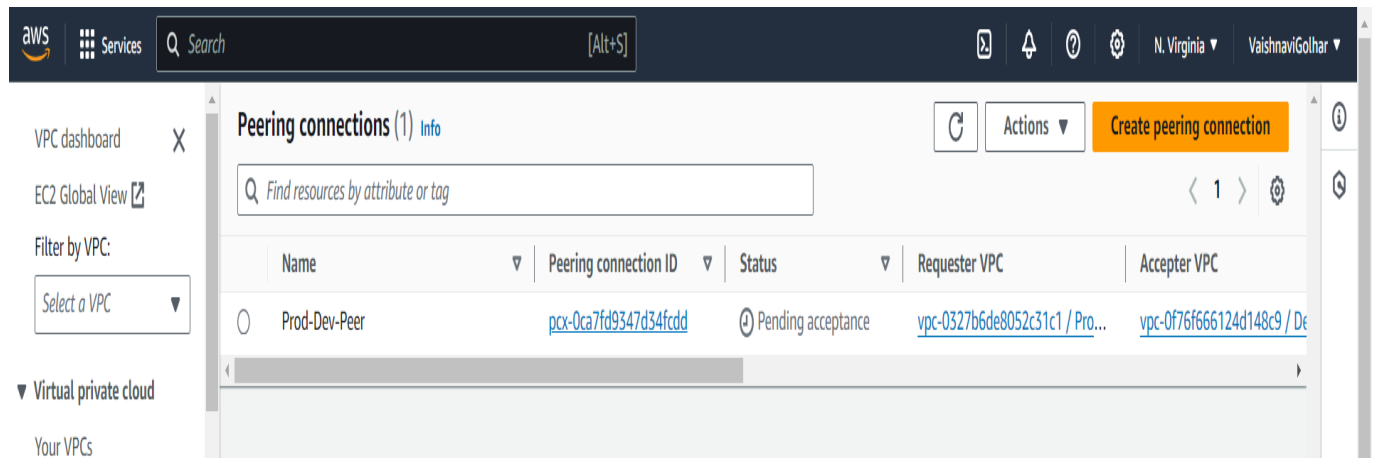
Add new tag

You can add 49 more tags.

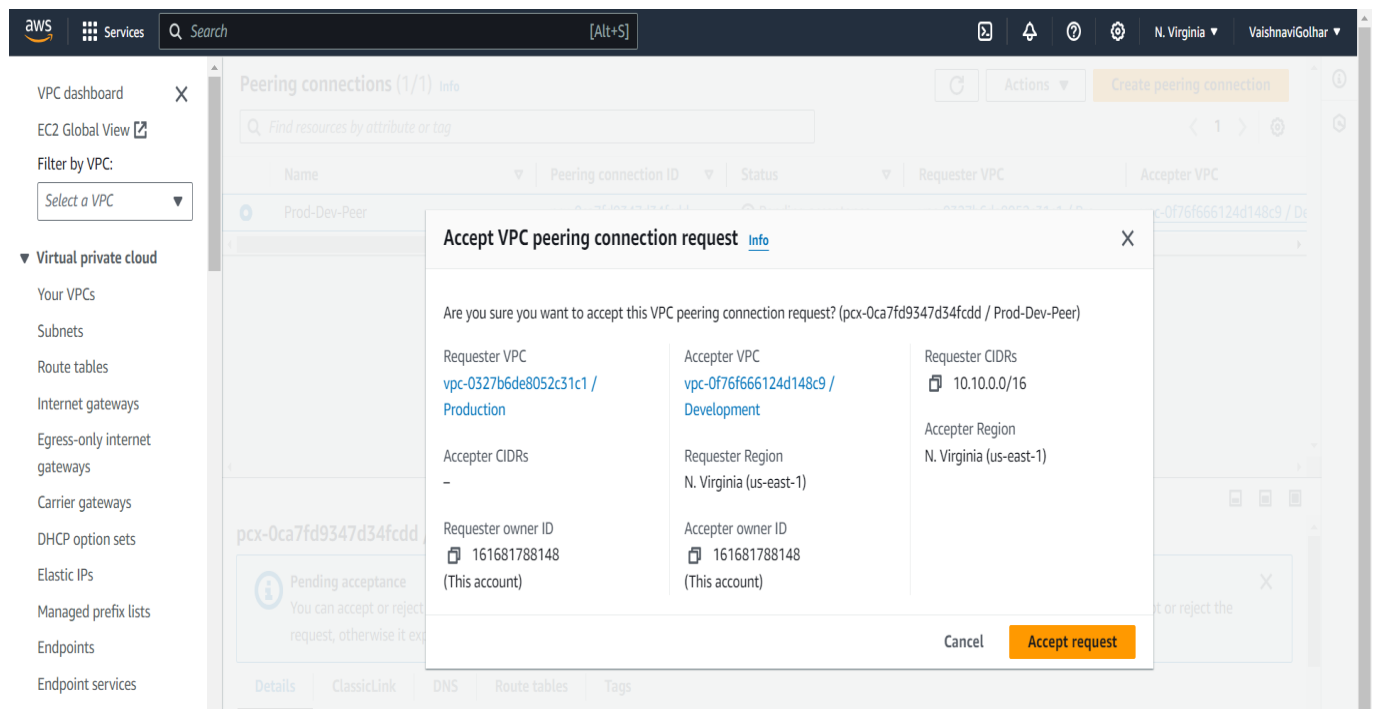
Cancel

Create peering connection

3. Peering connection is created with pending acceptance.



4. Go to action and accept it



Now peering connection is successful.

5. Go to route table and create route table with a name PROD-DEV-PEER-RT in the Production VPC.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

vpc-0327b6de8052c31c1 (Production) ▼

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q PROD-DEV-PEER-RT X

Remove

Add new tag

You can add 49 more tags.

Cancel **Create route table**

6. Edit route give the CIDR of DB-2 (20.20.2.0/24) instance of the Development subnet.

aws Services Search [Alt+S]

VPC > Route tables > rtb-0f2e632bf3525901a > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No
Q 20.20.2.0/24 X	Peering Connection	-	No
	Q pcx- X		
	Use: "pcx-"		
	pcx-0ca7fd9347d34fcdd (Prod-Dev-Peer)		

Add route

Cancel Preview **Save changes**

7. Make subnet association with DB (10.10.4.0/24) instance of the Production Subnet.

aws

Services

Search

[Alt+S]

N. Virginia

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VPC

>

Route tables

>

rtb-0f2e632bf3525901a

>

Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/5)

Filter subnet associations

< 1 > ⚙

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	APP-2	subnet-04d7e87751d2ac3b6	10.10.2.0/24	–	Main (rtb-0ab4a9b54f36d8c2e)
☐	DB-CACHE	subnet-0d5b55b68a859d099	10.10.3.0/24	–	rtb-026df6ed92c83c983 / NAT-PRODU..
☒	DB	subnet-0c799334c98542344	10.10.4.0/24	–	rtb-0f2e632bf3525901a / PROD-DEV...
☐	WEB	subnet-08ea3d1cf9db288c3	10.10.5.0/24	–	rtb-0e9eeae311bd854a8 / Production...
☐	APP-1	subnet-059eea9c4dfd9e3a7	10.10.0.0/24	–	rtb-026df6ed92c83c983 / NAT-PRODU..

Selected subnets

subnet-0c799334c98542344 / DB X

Cancel Save associations

8. Similarly create another route table with a name DEV-PROD-Peer-RT in the Development VPC.

Create route table

Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

Dev-PROD-Peer-RT

VPC

The VPC to use for this route table.

vpc-0f76f666124d148c9 (Development)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q Dev-PROD-Peer-RT X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

9. Edit route give the CIDR of the DB (1010.4.0/24) instance of the Production subnet.

aws Services Search [Alt+S] N. Virginia VaishnaviGolhar

VPC > Route tables > rtb-0b03bf5376c6fc75d > Edit routes

Edit routes

Destination	Target	Status	Propagated
20.20.0.0/16	local	Active	No
10.10.4.0/24	pcx-0ca7fd9347d34fcdd	-	No

Remove

Add route

Cancel Preview Save changes

10. Make subnet association with DB-2 (20.20.2.0/24) instance of the Development subnet.

aws Services Search [Alt+S] N. Virginia VaishnaviGolhar

VPC > Route tables > rtb-0b03bf5376c6fc75d > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	WEB-2	subnet-094d9e6451353274e	20.20.1.0/24	-	rtb-02078046199a48905 / Developme..
☒	DB-2	subnet-0f3679750b5c0f581	20.20.2.0/24	-	Main (rtb-08deae1385c2bedd2)

Selected subnets

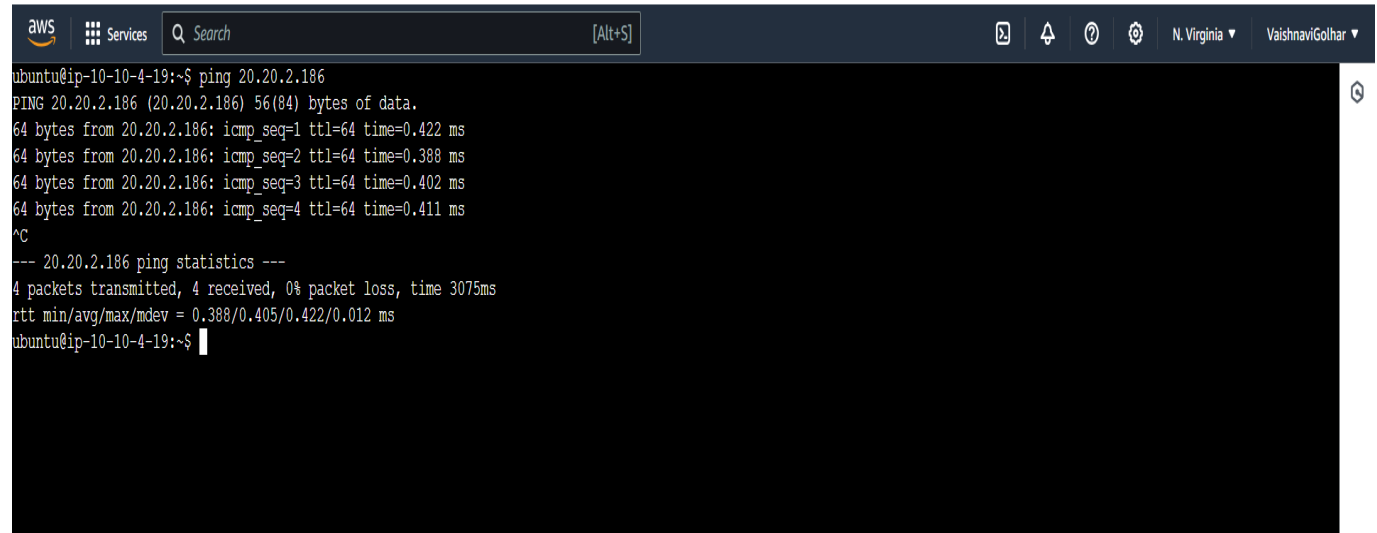
subnet-0f3679750b5c0f581 / DB-2

Cancel Save associations

11. Two way peering connection is done.

12. Now connect WEB instance ubuntu of Production to DB instance.

13. Now ping from DB instance of Production Network to DB-2 instance of Development Network.



```
aws
Services
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ubuntu@ip-10-10-4-19:~$ ping 20.20.2.186
PING 20.20.2.186 (20.20.2.186) 56(84) bytes of data.
64 bytes from 20.20.2.186: icmp_seq=1 ttl=64 time=0.422 ms
64 bytes from 20.20.2.186: icmp_seq=2 ttl=64 time=0.388 ms
64 bytes from 20.20.2.186: icmp_seq=3 ttl=64 time=0.402 ms
64 bytes from 20.20.2.186: icmp_seq=4 ttl=64 time=0.411 ms
^C
--- 20.20.2.186 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3075ms
rtt min/avg/max/mdev = 0.388/0.405/0.422/0.012 ms
ubuntu@ip-10-10-4-19:~$
```

